

Chapter 5

Network Layer:

The Control Plane

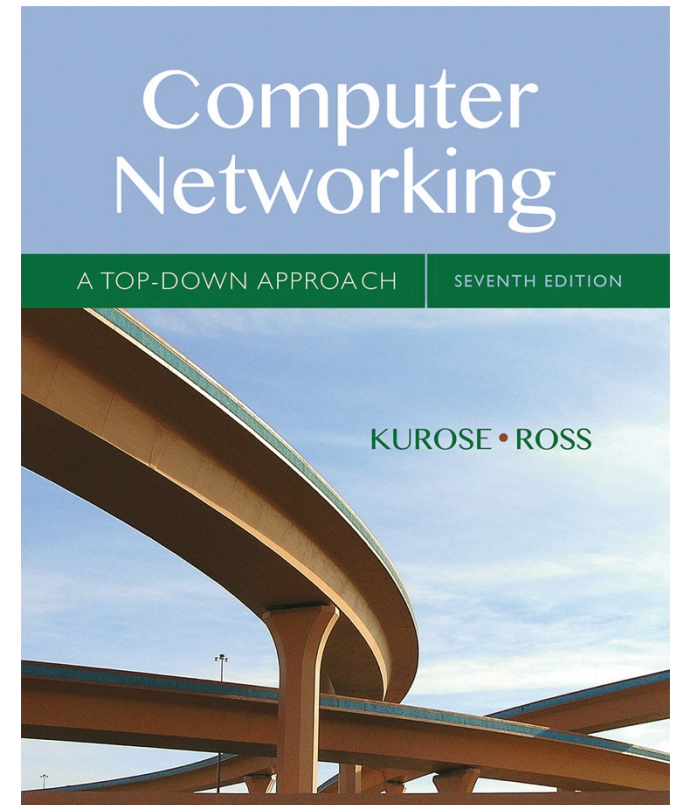
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Computer Networking: A Top Down Approach

7th edition

Jim Kurose, Keith Ross

Pearson/Addison Wesley

April 2016

Chapter 5: network layer control plane

understand principles behind network control plane

- traditional routing algorithms

and their instantiation, implementation in the Internet:

- OSPF, BGP

Chapter 5: outline

5.1 introduction

- Control Plane
- Autonomous Systems

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

- RIP
- OSPF

5.4 inter-AS routing in the Internet: B

Network-layer functions

Recall: two network-layer functions:

- *forwarding*: move packets from router's input to appropriate router output

data plane

- *routing*: determine route taken by packets from source to destination

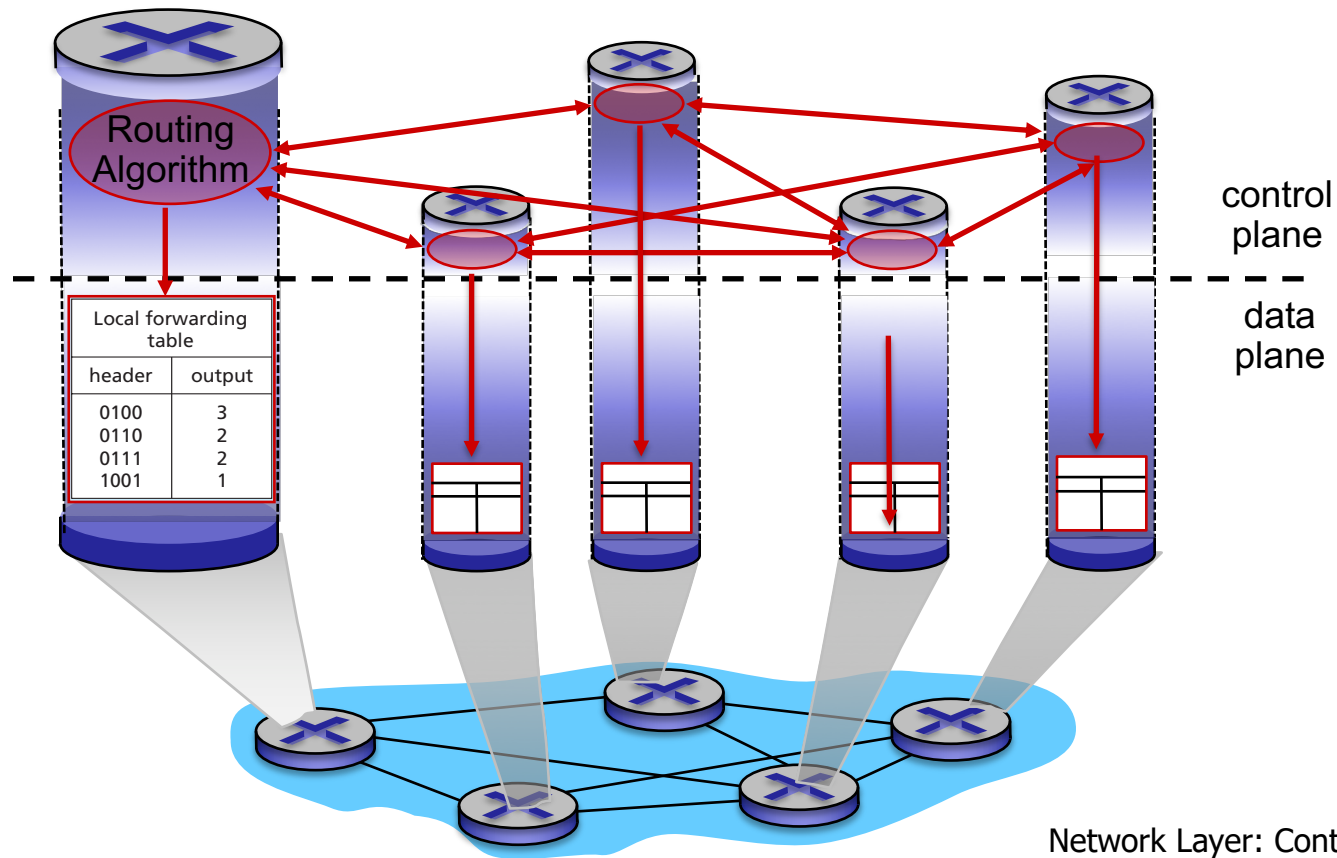
control plane

Two approaches to structuring network control plane:

- per-router control (traditional)
- logically centralized control (software defined networking)

Per-router control plane

Individual routing algorithm components *in each and every router* interact with each other in control plane to compute forwarding tables



Making routing scalable

our routing study thus far - idealized

- all routers identical
- network “flat”

... *not* true in practice

scale: with billions of destinations:

- can't store all destinations in routing tables!
- routing table exchange would swamp links!

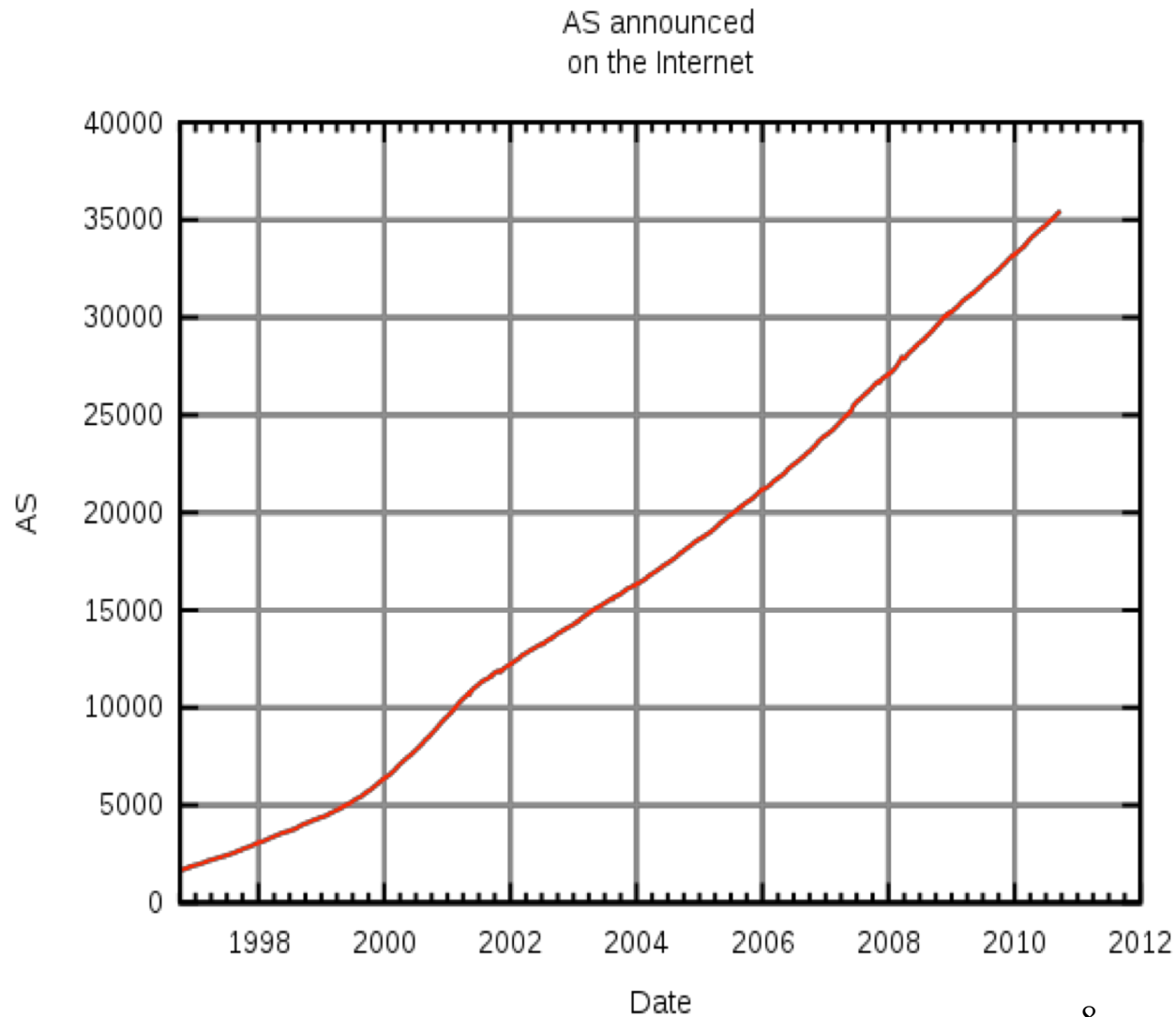
administrative autonomy

- internet = network of networks
- each network admin may want to control routing in its own network

Autonomous Systems

- an **autonomous system (AS)** is a region of the Internet that is administered by a single entity and that has a unified routing policy
- each autonomous system is assigned an Autonomous System Number (**ASN**). Each ASN is either 16bits or 32bits
 - ASN assigned by Regional Internet Registries
 - some are reserved for private use and never appear on the Internet
 - example ASNs
 - U of T's campus network (AS239)
 - Sprint (AS1239, AS1240, AS 6211, ...)

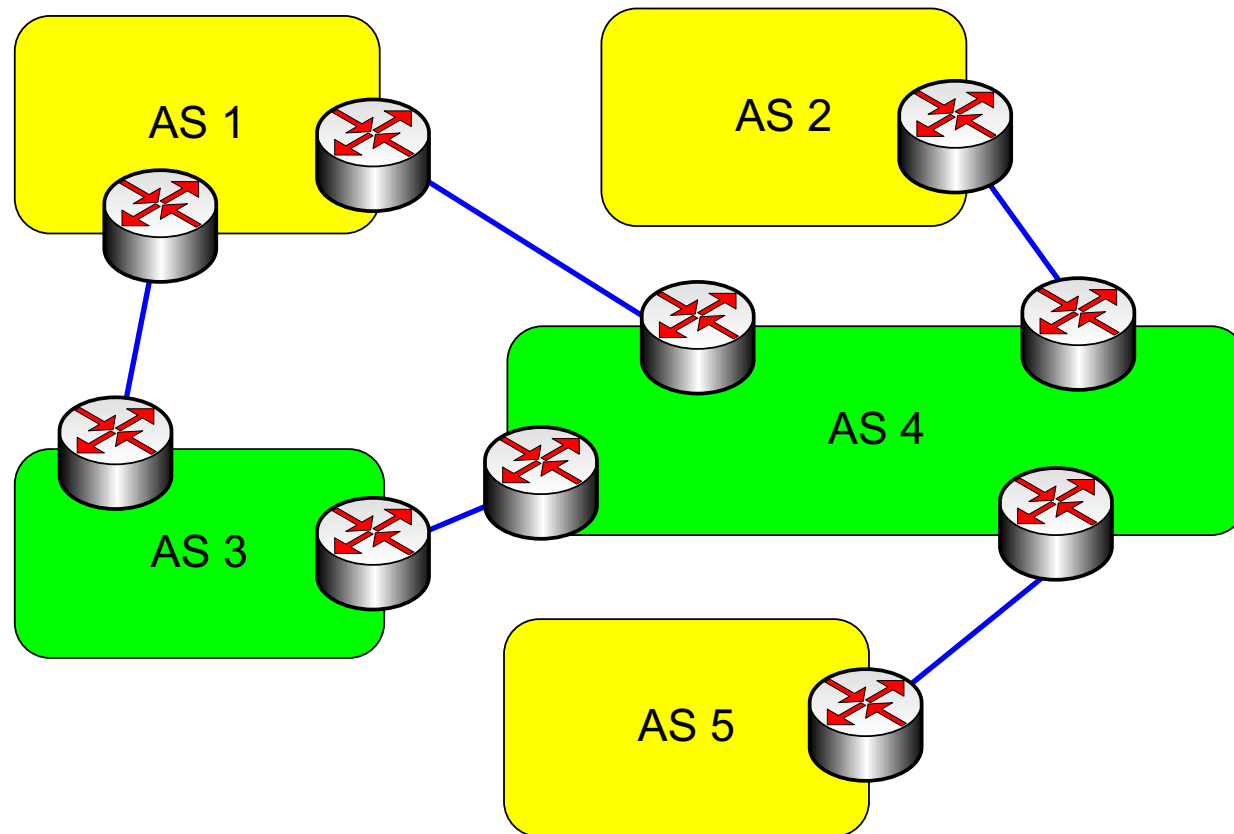
Number of Autonomous Systems



Autonomous Systems terminology

- **local traffic**: traffic with source and destination in AS
- **transit traffic**: traffic that passes through the AS
- **Stub AS**: has connection to only one AS, only carries local traffic
- **Multihomed Stub AS**: has connection to >1 AS, but does not carry transit traffic
- **Transit AS**: has connection to >1 AS and carries transit traffic

Stub and Transit Networks



- AS 1 is a **multi-homed stub** network
- AS 3 and AS 4 are **transit** networks
- AS 2 and AS 5 are **stub** networks

Internet approach to scalable routing

aggregate routers into regions known as “**autonomous systems**” (AS) (a.k.a. “domains”)

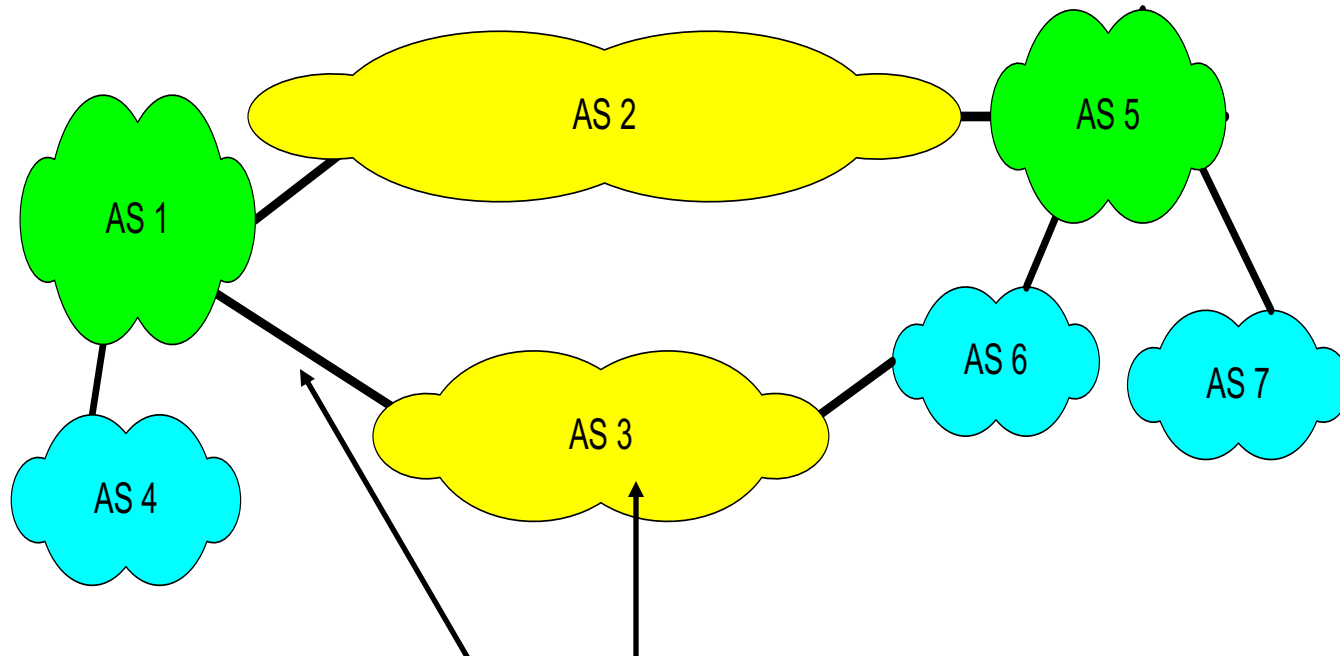
intra-AS routing

- routing among hosts, routers in same AS (“network”)
- all routers in AS must run *same* intra-domain protocol
- routers in *different* AS can run *different* intra-domain routing protocol
- gateway router: at “edge” of its own AS, has link(s) to router(s) in other AS'es

inter-AS routing

- routing among AS'es
- gateways perform inter-domain routing (as well as intra-domain routing)

Interdomain and Intradomain Routing



- routing protocols used **inside** an AS, referred to as intradomain routing, are called interior gateway protocols (IGP)
 - objective: shortest path, only operate within an AS
- routing protocols used **between** ASs, referred to as interdomain routing, are called exterior gateway protocols (EGP)
 - objective: satisfy policy of the ASs, not always shortest path

Interdomain and Intradomain Routing

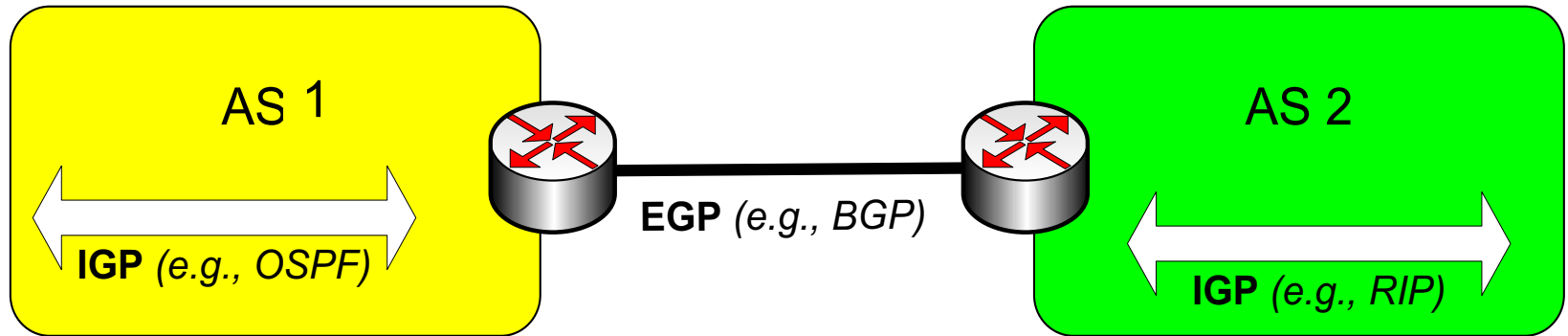
Intradomain Routing

- routing within an Autonomous System (AS)
- ignores the Internet outside the AS
- protocols for Intradomain routing are collectively called **Interior Gateway Protocols** or **IGP's**.
- popular protocols are:
 - RIP (simple, old)
 - OSPF (better)

Interdomain Routing

- routing between AS's
- assumes that the Internet consists of a collection of interconnected AS's
- normally, there is one dedicated router in each AS that handles interdomain traffic.
- protocols are collectively called **Exterior Gateway Protocols** or **EGP's**.
- popular protocols are:
 - Border Gateway Protocol (BGP) v4 current

EGP and IGP



- Interior Gateway Protocol (IGP)
 - routing is done based on **metrics**
 - routing domain is **one AS**
- Exterior Gateway Protocol (EGP)
 - routing is done based on **policies**
 - routing domain is the **entire Internet**